

Experiment 3: Average Sale Price of Houses

AIM

To find the average sale price of houses having more than four bedrooms using NumPy arrays and operations

ALGORITHM

1. Import NumPy.
2. Create a NumPy array containing the number of bedrooms, square footage and sale price.
3. Select houses having more than four bedrooms.
4. Extract the sale prices of the selected houses.
5. Calculate their average sale price using `np.mean()`.
6. Display the result.

CODE

```
import numpy as np

house_data = np.array([

    [3, 1500, 250000],

    [5, 2200, 400000],

    [6, 2500, 500000],

    [4, 1800, 300000],

    [7, 3000, 600000]

])

houses_more_than_4 = house_data[house_data[:, 0] > 4]

sale_prices = houses_more_than_4[:, 2]

average_price = np.mean(sale_prices)

print("Houses with more than 4 bedrooms:")

print(houses_more_than_4)

print("\nAverage sale price:", average_price)
```

INPUT

Bedrooms Square Feet Sale Price

3	1500	250000
5	2200	400000
6	2500	500000
4	1800	300000
7	3000	600000

OUTPUT

```
import numpy as np
house_data = np.array([
    [3, 1500, 250000],
    [5, 2200, 400000],
    [6, 2500, 500000],
    [4, 1800, 300000],
    [7, 3000, 600000]
])
houses_more_than_4 = house_data[house_data[:, 0] > 4]
sale_prices = houses_more_than_4[:, 2]
average_price = np.mean(sale_prices)
print("Houses with more than 4 bedrooms:")
print(houses_more_than_4)
print("\nAverage sale price:", average_price)
```

```
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:1
3:20) [MSC v.1940 32 bit (Intel)] on win32
Type "help", "copyright", "credits" or "license()" for
more information.
>>>
= RESTART: C:/Users/monik/Desktop/SLOT B & C/FODS/expl
.py
Houses with more than 4 bedrooms:
[[ 5 2200 400000]
 [ 6 2500 500000]
 [ 7 3000 600000]]

Average sale price: 500000.0
>>>
```

RESULT

Thus, the average sale price of houses having more than four bedrooms was calculated successfully as **500000**.